

**NEW STARS**

At the top of this image are the Trapezium stars forming within the Orion Nebula. Also visible, toward the bottom left-hand corner, is a line of shock waves created by material outflowing from the embryonic stars at a speed of 450,000 mph (720,000 kph).

EMISSION NEBULA**Orion Nebula**

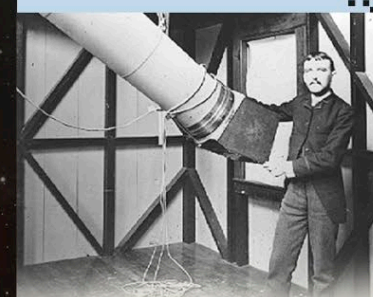
CATALOG NUMBERS
M42, NGC 1976

DISTANCE FROM SUN
1,500 light-years

MAGNITUDE 4

ORION

The most famous and the brightest nebula in the night sky, the Orion Nebula is easily visible with the naked eye as a diffuse, reddish patch below Orion's belt (see pp.390–391). It is also the closest emission nebula to Earth and has been extensively studied. The nebula spans about 30 light-years and has an apparent diameter four times that of a full moon. However, it is a small part of a much larger molecular cloud system known as OMC-1, which has a diameter of several hundred light-years. The Orion Nebula sits at the edge of OMC-1, which stretches as far as the Horsehead Nebula (opposite). The nebula glows with the ultraviolet radiation of the new stars forming within it. Many of these stars have been shown to have protoplanetary disks surrounding them. The principal stars whose radiation is ionizing the cloud of dust and gas belong to the Trapezium star cluster (see p.391), located at the heart of the nebula. At about 30,000 years old, the Trapezium is one of the youngest clusters known. It is a quadruple star system consisting of hot OB stars (see pp.232–233). In 1967, an extended dusty region was discovered directly behind the Orion Nebula. Known as the Kleinmann–Low Nebula, it has strong sources of infrared radiation embedded within it. These sources are believed to be protostars and newly formed stars.

EXPLORING SPACE**FIRST PHOTOGRAPH**

A pioneer of astrophotography, the American scientist Henry Draper (1837–1882) took the first photo of a nebula in September 1870 after he turned his camera to the Orion Nebula, the brightest one in the sky. Although his photograph was relatively crude, 12 years later, he used an 11-in (28-cm) photographic refractor to obtain a much-improved image. The Orion Nebula has since been photographed probably more times than any other nebula.

THE GREAT ORION NEBULA

This view was captured with the VISTA telescope in Chile. It is an infrared image, revealing newborn stars within the nebula's dusty interior.